# Singapore School Kinderland

## Secondary 1 (Sec 1) Computer Science — Term 2 Comprehensive Practice Worksheet

*Topics: Error Detection Methods (TB Pg 54–62) & Encryption (TB Pg 63–74) | Total Marks: 40*

**Student Name:** _____ **Class:** Sec 1 | **Date:** _____ **Marks:** / 40

## SECTION A: PARITY CHECKS & PARITY BLOCKS (10 MARKS)

**Q1. Parity Bit Calculation** *[4 Marks]*

A computer system transmits 7-bit ASCII data. Add the required Parity Bit (most significant bit) for each byte below:

| Byte | 7-bit Data Payload | Parity System | Required Parity Bit (0 or 1) | Complete 8-bit Transmitted Byte |
|---|---|---|---|---|
| A | 1 1 0 0 1 0 1 | Even Parity | | |
| B | 1 0 1 1 1 1 0 | Even Parity | | |
| C | 0 1 1 0 0 0 1 | Odd Parity | | |
| D | 1 1 1 0 1 1 1 | Odd Parity | | |

**Q2. Parity Block Matrix Analysis & Auto-Correction** *[6 Marks]*

A block of four bytes was transmitted using **Even Parity**. An extra parity byte was appended to the bottom:

| Row | Bit 7 (Parity) | Bit 6 | Bit 5 | Bit 4 | Bit 3 | Bit 2 | Bit 1 | Bit 0 | Row Status (OK / Error) |
|---|---|---|---|---|---|---|---|---|---|
| Byte 1 | 1 | 0 | 1 | 1 | 0 | 0 | 1 | 0 | |
| Byte 2 | 0 | 1 | 1 | 0 | 1 | 0 | 1 | 1 | |
| Byte 3 | 1 | 1 | 0 | 1 | 1 | 0 | 0 | 1 | |
| Byte 4 | 1 | 0 | 0 | 1 | 0 | 1 | 1 | 0 | |
| **Parity Byte** | **1** | **0** | **0** | **1** | **0** | **1** | **1** | **0** | |

(a) Identify the corrupted row and column: Corrupted Row: _____ | Corrupted Column: _____ *[2]*

(b) State the original corrupted bit value: _____ and state the corrected bit value: _____ *[2]*

(c) Explain why this parity block cannot correct an error if two bits flip in the same row and column: *[2]*

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## SECTION B: CHECKSUMS, ECHO CHECKS & CHECK DIGITS (10 MARKS)

**Q3. Checksum Verification Process** *[3 Marks]*

Describe how a checksum algorithm ensures that a transmitted data file has arrived without corruption.

**Q4. Echo Check Critique** *[3 Marks]*

A student proposes using an Echo Check for online multiplayer gaming. Give TWO distinct reasons why an Echo Check is unsuitable for high-speed network applications.

Reason 1:

Reason 2:

**Q5. Check Digits vs Transmission Checks** *[4 Marks]*

(a) State TWO human data entry errors that check digits detect but parity checks do not. *[2]*

(b) The following 12 digits of an ISBN-13 barcode are scanned: `9 7 8 1 5 1 0 4 8 4 4 2`.
Calculate the final 13th Check Digit using the weighting `(1, 3, 1, 3, 1, 3, 1, 3, 1, 3, 1, 3)`. Show your calculation: *[2]*

## SECTION C: ARQ PROTOCOL SCENARIOS & TIMEOUTS (10 MARKS)

**Q6. ARQ Protocol Operation** *[5 Marks]*

The diagram below outlines an Automatic Repeat Request (ARQ) transmission sequence between Computer A and Computer B:

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Step 1: Computer A sends Packet 1 → starts Timeout Timer.
Step 2: Computer B receives Packet 1 → calculates Checksum → Checksum matches.
Step 3: Computer B sends [ Message X ] → Computer A cancels timer → sends Packet 2.
Step 4: Packet 2 gets corrupted by electrical noise in transit.
Step 5: Computer B detects checksum mismatch → sends [ Message Y ].
Step 6: Computer A receives Message Y → immediately re-transmits [ Packet Z ].
```

(a) Identify Message X: _____ *[1]*

(b) Identify Message Y: _____ *[1]*

(c) Identify Packet Z: _____ *[1]*

(d) Explain what Computer A will do if Packet 1 was completely dropped by a faulty router and Computer B received nothing: *[2]*

**Q7. ARQ Timeout Timer Evaluation** *[5 Marks]*

(a) Explain what would happen if the sender's timeout timer was configured to be **too short**: *[2.5]*

(b) Explain what would happen if the sender's timeout timer was configured to be **too long**: *[2.5]*

## SECTION D: ENCRYPTION, KEYS & DIGITAL CERTIFICATES (10 MARKS)

**Q8. Symmetric vs Asymmetric Encryption** *[5 Marks]*

(a) Explain what is meant by the **Key Distribution Problem** in symmetric encryption: *[2]*

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(b) Complete the comparison table below: *[3]*

| Feature | Symmetric Encryption | Asymmetric Encryption |
|---|---|---|
| Number of keys required per user pair | | |
| Key used to decrypt message | | |
| Relative processing speed (Fast / Slow) | | |

**Q9. Asymmetric Key Exchange & Digital Certificates** *[5 Marks]*

A student named David wants to purchase books from an online store securely.

(a) Which key will David's browser use to encrypt his credit card number? *[1]*
Answer: _____

(b) Which key will the online store use to decrypt David's credit card number? *[1]*
Answer: _____

(c) State TWO items contained within the store's SSL/TLS Digital Certificate that confirm its authenticity: *[2]*

Item 1: _____

Item 2: _____

(d) State the role of the Certificate Authority (CA): *[1]*

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